UNIT – 3
Network Layer

What is Network Layer?

- The basic function of **Network layer is to provide an end to end communication** capability to the transport layer which lies above it at sender level similarly receiver level.
- The **network layer holds the responsibility of managing subnet performance.**
- **This layer is more focused to control the operations of data transmission, routing and switching technologies, packet forwarding and sequencing.**
- To achieve the goal the network layer must know the **Topology of the communication Subnet** i.e **set of all routers and choose appropriate path through it.**
- The **Network layer protocols are concern exchange of packets of information** between transport layer entities.
- A **packet is a group of bits (i.e collection of frames).**
- The **services provided by the network layer to transport layer is called Network services.**
- It **makes routing of data through network from source to destination**.

The **Network Layer** is the **3rd layer of the OSI model**. It is responsible for delivering packets from the **source device to the destination device**, even when they are on different networks.

Main Points

1.Logical Addressing

1. Uses **IP addresses** to identify devices.
2. Example: `192.168.1.10`

2.Routing

1. Determines the **best path** for packets to travel from source to destination.

3.Packet Forwarding

1. Forwards packets from one network to another through routers.

4.Internetworking

1. Connects different networks and allows them to communicate.

5.Packetization

1. Receives data from the Transport Layer and divides it into **packets**.

6.Fragmentation

1. Breaks large packets into smaller pieces when required by the network

7. Routers

1. Routers mainly operate at the **Network Layer**.
2. They use IP addresses to forward packets.

8.**Network Layer Protocols**

1. **IPv4 ,IPv6,ICMP**

9.Path Selection

1. Selects a suitable route using routing algorithms and

Store and Forward Packet switching:

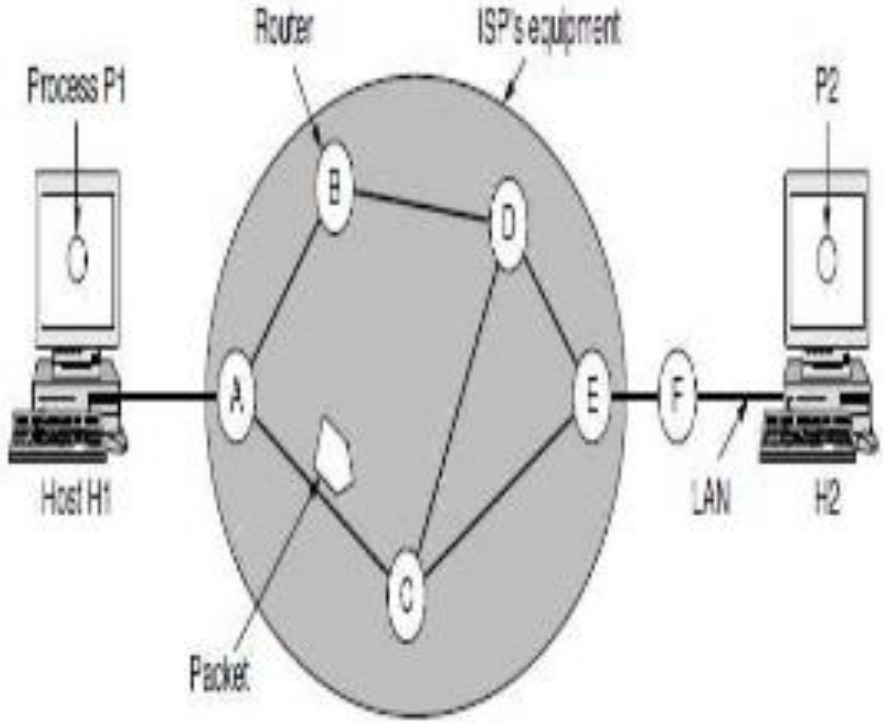
- **Packet switching is a method of transferring the data to a network in form of packets.**
- **In order to transfer the file fast** and **efficient manner over the network** and **minimize the transmission latency**(delay), **the data is broken into small pieces of variable length, called Packet**.
- **At the destination, all these small-parts (packets) has to be reassembled,** belonging to the same file.
- Packet Switching uses **Store and Forward technique** while switching the **packets**; while **forwarding the packet each hop first store that packet then forward.**
- This technique is very beneficial because packets may get discarded at any hop due to some reason.
- **Each router verifies possible number of links which link is better.**
- **More than one path is possible between a pair of source and destination.**
- **Each packet contains Source and destination address** using which they **independently travel through the network.**
- In other words, **packets belonging to the same file may or may not travel through the same path.**

Store-and-forward packet switching is a method in which a network device **receives the complete packet, stores it temporarily, and then forwards it to the next device**.

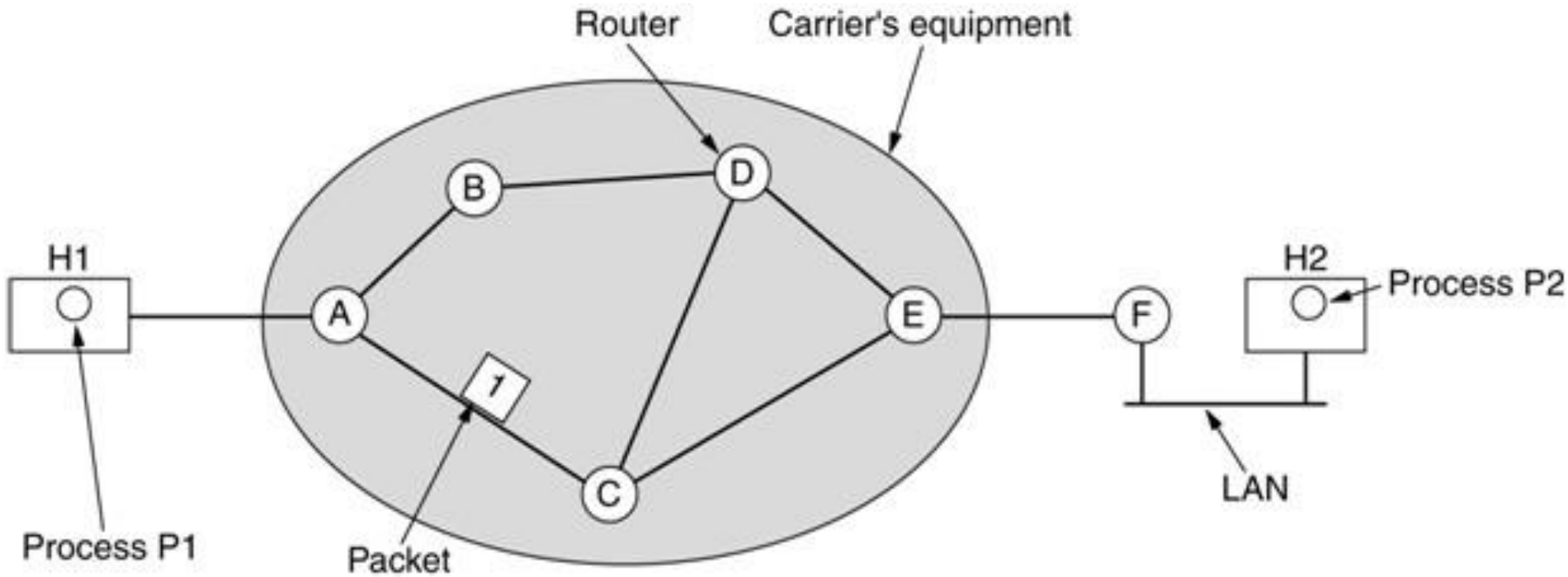
Working Principle:

1.Data is divided into packets.
2.The packet is sent from the **source** to a router/switch.
3.The router **stores the complete packet** in its memory.
4.It checks the packet and determines the **next destination/path**.
5.The router **forwards the packet** to the next router.
6.This process continues until the packet reaches the **destination**.

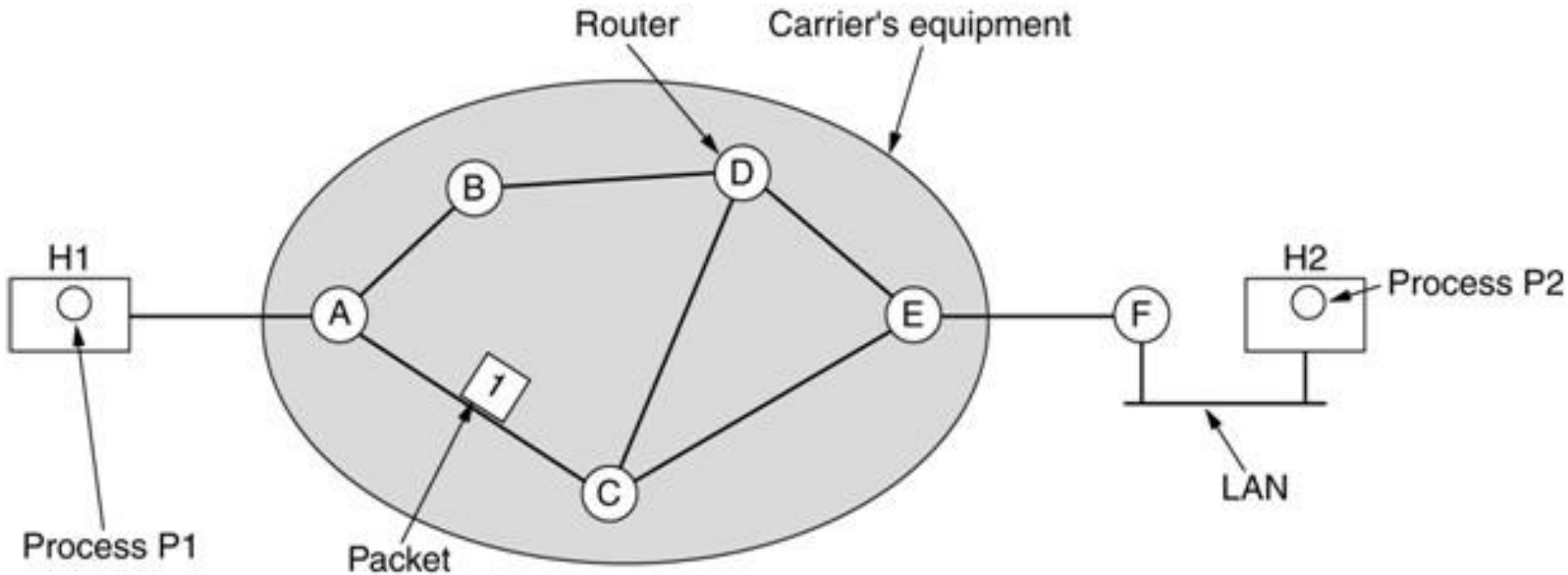
Store-and-Forward Packet Switching



- A host with a packet to send transmits it to the nearest router.
- The packet is stored there until it has fully arrived.
- the link has finished its processing by verifying the checksum.
- Then it is forwarded to the next router along the path until it reaches the destination host.
- This mechanism is store-and-forward packet switching.


Router
Carrier's equipment
B
D
H1
A
E
F
H2
Process P2
1
C
LAN
Process P1
Packet

- Host H1 is directly connected to one of the carrier's routers, A, by a leased line.
- In contrast, H2 is on a LAN with a router, F, owned and operated by the customer.
- This router also has a leased line to the carrier's equipment. We have shown F as being outside the oval because it does not belong to the carrier, but in terms of construction, software, and protocols, it is probably no different from the carrier's routers.
- Whether it belongs to the subnet is arguable, routers on customer premises are considered part of the subnet because they run the same algorithms as the carrier's routers (and our main concern here is algorithms).



Services Provided to the Transport Layer:

- The **network layer provides services to the transport layer at the network layer/transport layer interface.**
- An important question is what kind of services the network layer provides to the transport layer.
- The network layer services have been designed with the following goals in mind.

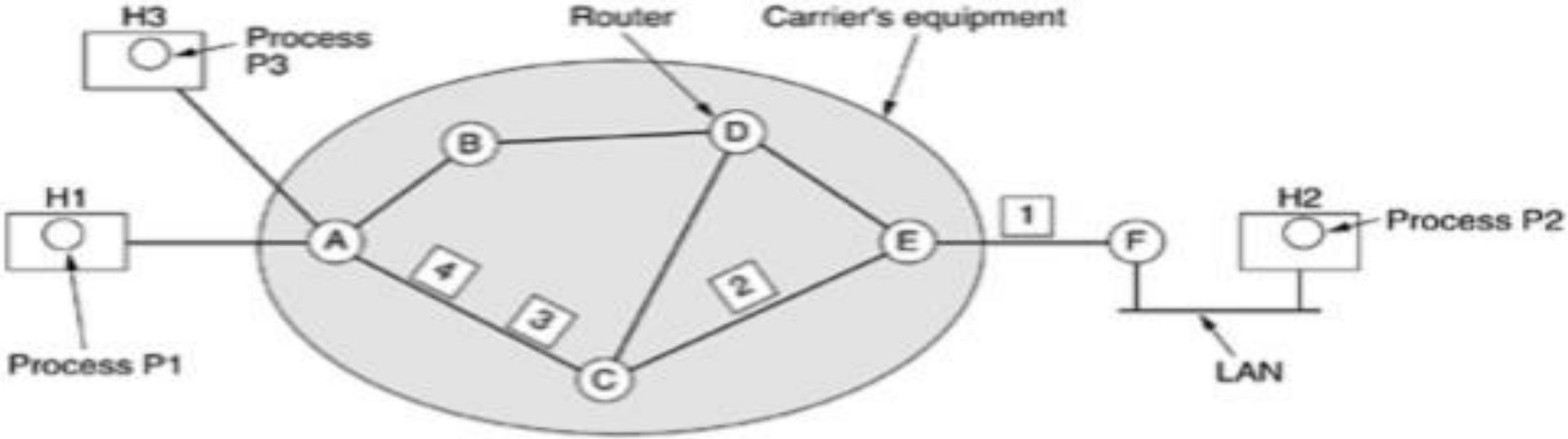
1. **The services should be independent of the router technology.**
2. **The transport layer should be shielded from the number, type, and topology of the routers present.**
3. **The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.**

- Given these goals, the designers of the network layer have a lot of freedom in writing detailed specifications of the services to be offered to the transport layer.
- whether the network layer should provide **connection-oriented service or connectionless service**.

Implementation of Connection oriented Service:

- connection-oriented service, we need a **virtual-circuit** subnet.
- The **virtual circuits is to avoid having to choose a new route for every packet sent** .
- All the packets are transmitted depends on **same path.**
- Instead, **when a connection is established, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers**.
- **That route is used for all traffic flowing over the connection, exactly the same way that the telephone system works.**
- **When the connection is released, the virtual circuit is also terminated.**
- With connection-oriented service, each packet carries an identifier telling which virtual circuit it belongs to.

Figure 5-3. Routing within a virtual-circuit subnet.



A's table

In		Out	
H1	1	C	1
H3	1	C	2

C's table

In		Out	
A	1	E	1
A	2	E	2

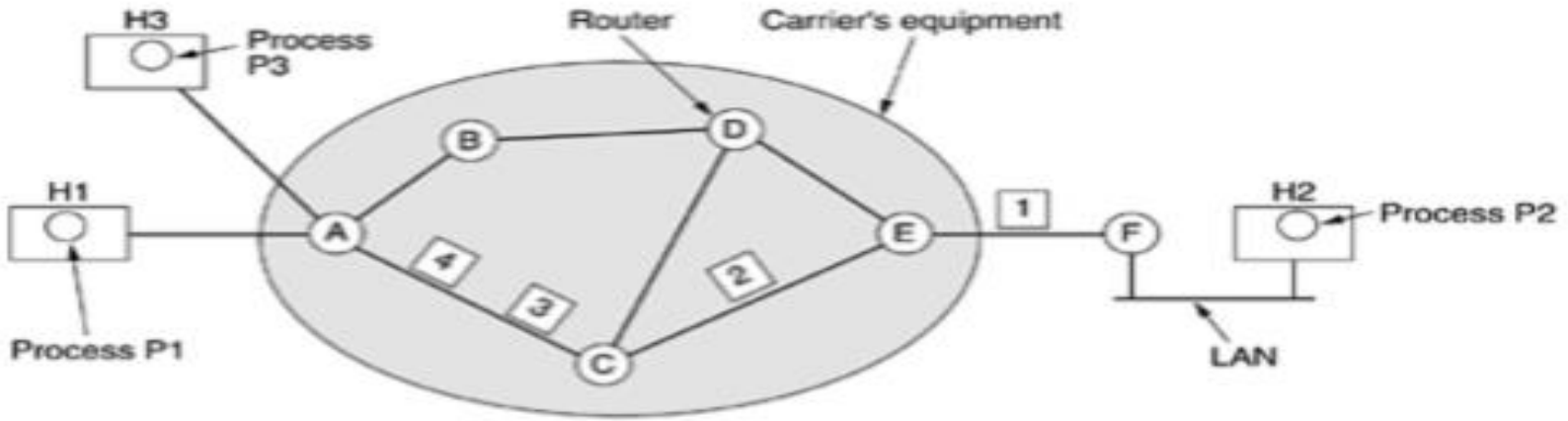
E's table

In		Out	
C	1	F	1
C	2	F	2

Example:

consider the situation of Fig. Here, host H1 has established connection 1 with host H2. It is remembered as the first entry in each of the routing tables. The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1. Similarly, the first entry at C routes the packet to E, also with connection identifier 1.

Figure 5-3. Routing within a virtual-circuit subnet.

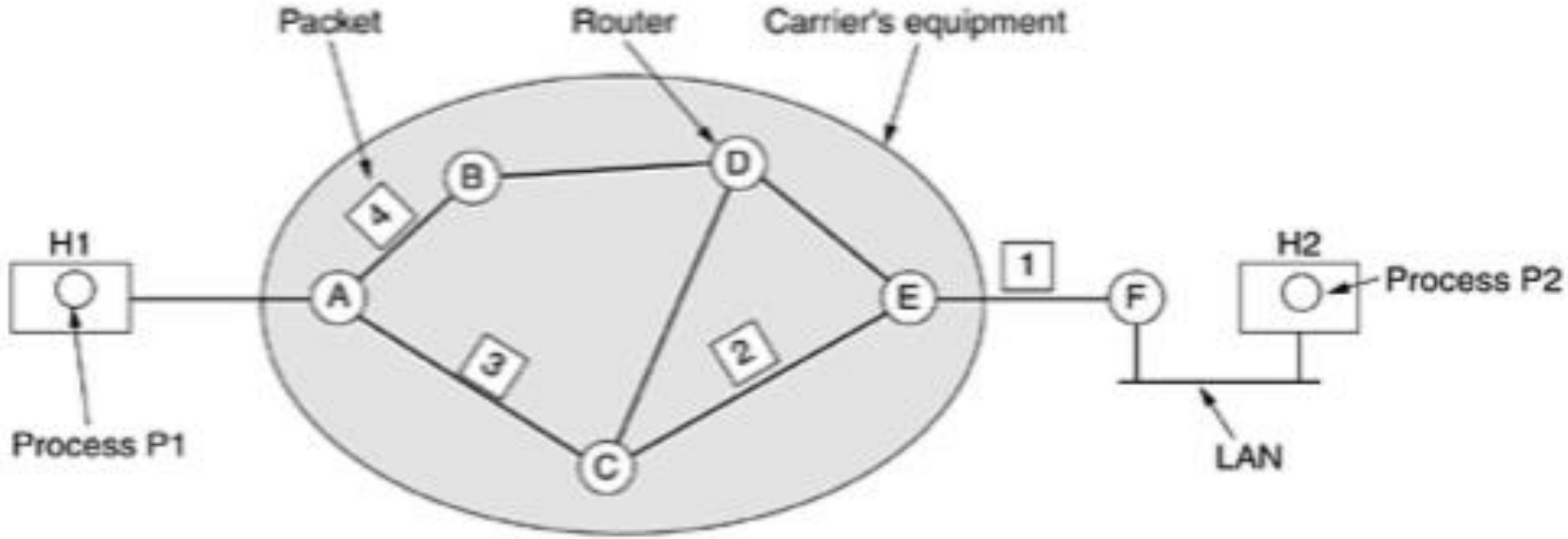


Now let us consider what happens if H3 also wants to establish a connection to H2. It chooses connection identifier 1 (because it is initiating the connection and this is its only connection) and tells the subnet to establish the virtual circuit. This leads to the second row in the tables. Note that we have a conflict here because although A can easily distinguish connection 1 packets from H1 from connection 1 packets from H3, C cannot do this. For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection. Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. In some contexts, this is called label switching.

Implementation of Connectionless Service:

- Connection less services is also called as **Datagrams**.
- **Datagram is a self contained message unit which contains sufficient information to allow it to be routed from source to destination.**
- Each packet is treated as independently.
- The network layer simply accepts messages from the transport layer and sends out to the destination.
- Datagram system is to postal system.
- **Each packet contains the full source and destination addresses .**
- **Each packet follows different routes from source to destination and they do not necessarily arrive in the same order. As** they transmitted i.e No sequence and flow control is done at receiver.

Figure 5-2. Routing within a datagram subnet.



A's table

initially		later	
A	–	A	–
B	B	B	B
C	C	C	C
D	B	D	B
E	C	E	B
F	C	F	B

Dest Line

C's table

A	A
B	A
C	–
D	D
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

Comparison of datagram and virtual-circuit subnets:

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Comparison of datagram and virtual-circuit networks

Routing Algorithms :

- **Routing** is process of establishing the routes that data packets must follow to reach the destination.
- In this process, a **routing table table is created which contains information regarding routes which data packets follow.**
- **Various routing algorithm are used for the purpose of deciding which route an incoming data packet needs to be transmitted on to reach destination efficiently.**

Classification of Routing Algorithms: The routing algorithms can be classified as follows:

1. **Adaptive Algorithms .**
2. **Non Adaptive Algorithms .**

1. Adaptive Algorithms :

- These are the algorithms which **change their routing decisions whenever network topology or traffic load changes.**
- The **changes in routing decisions are reflected in the topology as well as traffic of the network.**
- Also known as **dynamic routing**, these make use of dynamic information such as current topology, load, delay, etc. to select routes.
- **Optimization parameters are distance, number of hops and estimated transit time.**

2. Non-Adaptive Algorithms :

- These are the algorithms which **do not change t**heir routing decisions once they have been selected.
- This is also known as **static routing** as route to be taken is computed in advance and downloaded to routers when router is booted.

Optimality Principle:

Choosing the best path from the source to the destination by using optimality principle.

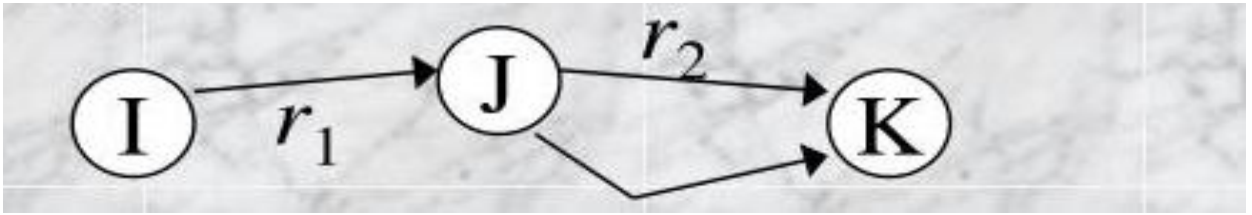
The set of optimal routes (best routes) from all sources to the destination from a tree is called sink tree.

Sink tree is a tree but not having loops.

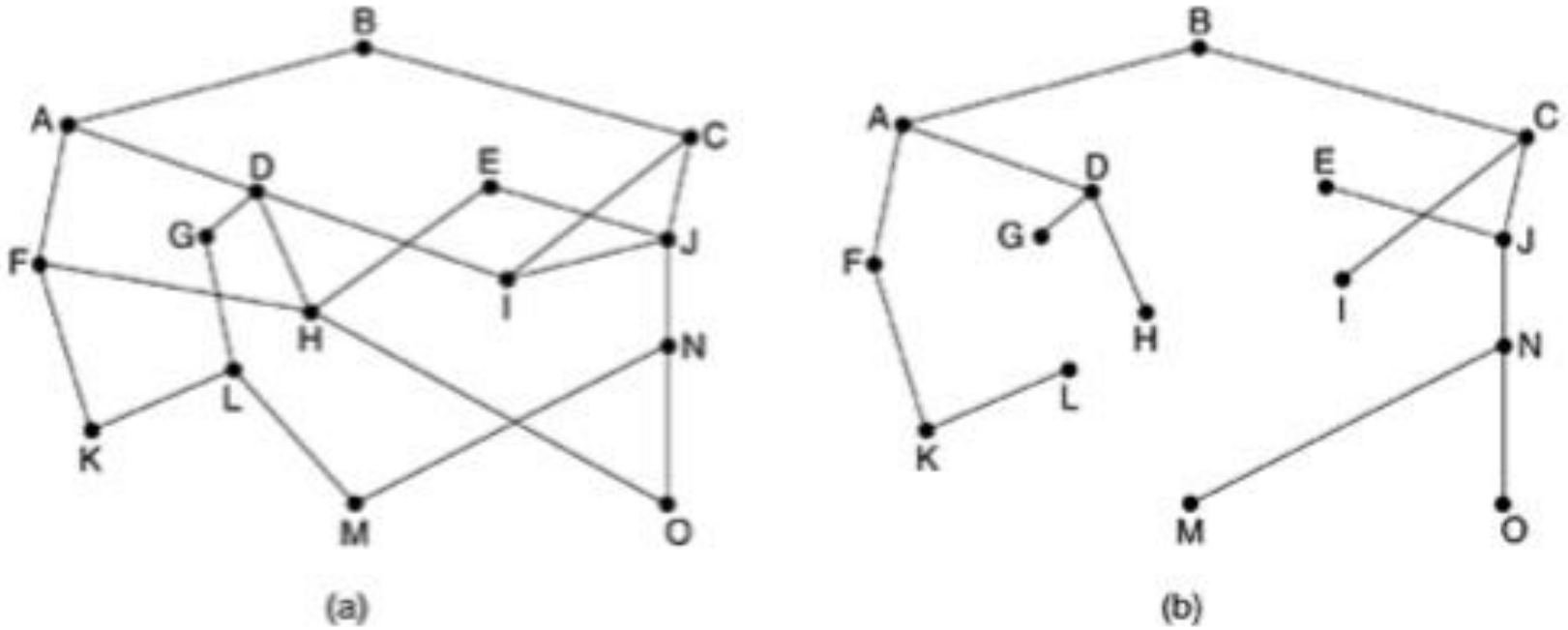
Sink trees are not unique may be different sink trees to construct in each tree.

Sink tree can cover all the routers (sources) in the subnet but not covering all the paths to reach the destination is called sink tree.

It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.



As a direct consequence of the optimality principle, we can see that the set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a **sink tree** and is illustrated in Fig., where the distance metric is the number of hops. **Note that a sink tree is not necessarily unique**; other trees with the same path lengths may exist. The goal of all routing algorithms is to discover and use the sink trees for all routers



(a) A subnet. (b) A sink tree for router **B.**

Optimality Principle:

Choosing the best path from the source to the destination by using optimality principle.

Sink tree can cover all the routers (sources) in the subnet but not covering all the paths to reach the destination is called sink tree.

Before we get into specific algorithms, it may be helpful to note that one can make a general statement about **optimal routes (best route) without regard to network topology or traffic**. This statement is known as the **optimality principle**.

It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.

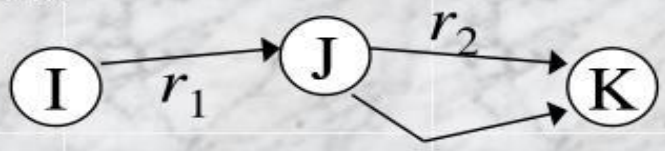
To see this, call the part of the route from I to Jr1 and the rest of the route r2. If a route better than r2 existed from J to K, it could be concatenated with r1 to improve the route from I to K, contradicting our statement that r1r2 is optimal.

Optimality Principle

- Optimality principle:
  - If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.
- This means:
  - Given the final destination, routers only need to know the optimal route to the next router.
- Construct a sink tree with the destination to be root.
  - The goal of all routing algorithms is to discover and use the sink tree for all routers.
  - Since it is a tree, there is no loops.
  - There would be different sink trees for each final destination.

The Optimality Principle

- If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.
  - the set of optimal routes from all sources to a destination form a tree, called a sink tree, rooted at the destination.



- The goal of all routing algorithms is to discover and use the sink trees for all routers.

Shortest Path Algorithm (Least cost Algorithm) : (Dijkstras Algorithm)

Several algorithms for **computing the shortest path between two nodes of a graph** are known. This one is due to **Dijkstras** (1959).

- **Each node is labeled (in parentheses) with its distance from the source node along the best known path**.
- **Initially, no paths are known, so all nodes are labeled with infinity.**
- **As the algorithm proceeds and paths are found, the labels may change, reflecting better paths.**
- **A label may be either tentative or permanent.**
- **Initially, all labels are tentative**. When it is discovered that a label represents the shortest possible path from the source to that node, it is made permanent and never changed thereafter.

Shortest Path Algorithm (Least cost Algorithm) : (Dijkstras)

Routing algorithms with a simple technique for computing optimal paths which gives a complete picture of the network and are able to form distributed routing algorithm to find, even though not all routers may know all of the details of the network.

The idea is to build a graph of the network, with **each node of the graph representing a router and each edge of the graph representing a communication line, or link.** To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.

- **In shortest path routing algorithm the path length between each node is measured either in Distance , Bandwidth , Average Traffic , Communication Cost or Delay** etc…
- This **Algorithm computes shortest path** according to any one of a number of criteria or a combination of criteria.
- **For this a Graph of subnet is drawn** . With **each node of a graph is representing a router , and each Arc of a graph is representing a communication link. Each link Associated with cost.**
- **To choose route b/w a given pair of routers , this algorithm's just find shortest path b/w them on the graph.**

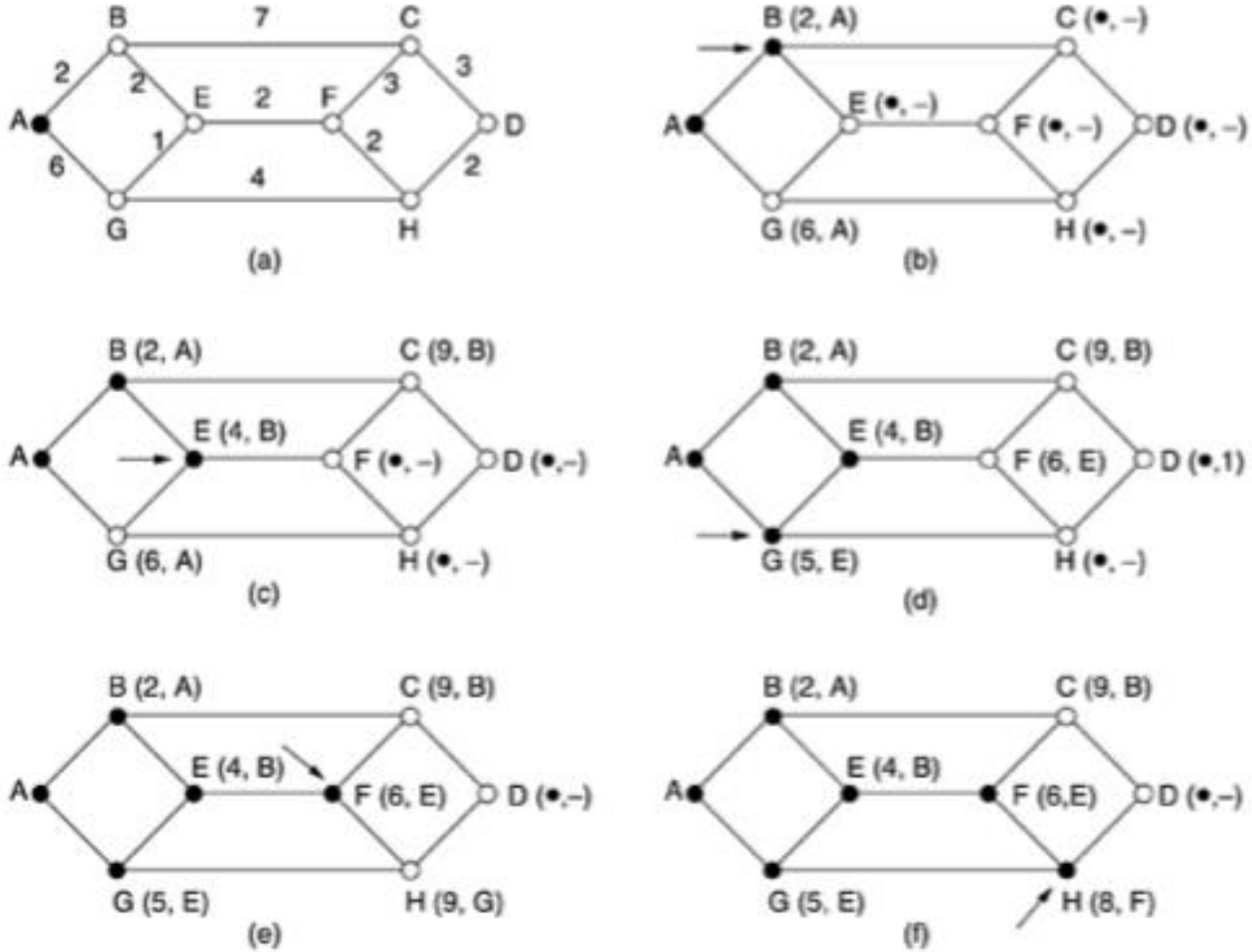
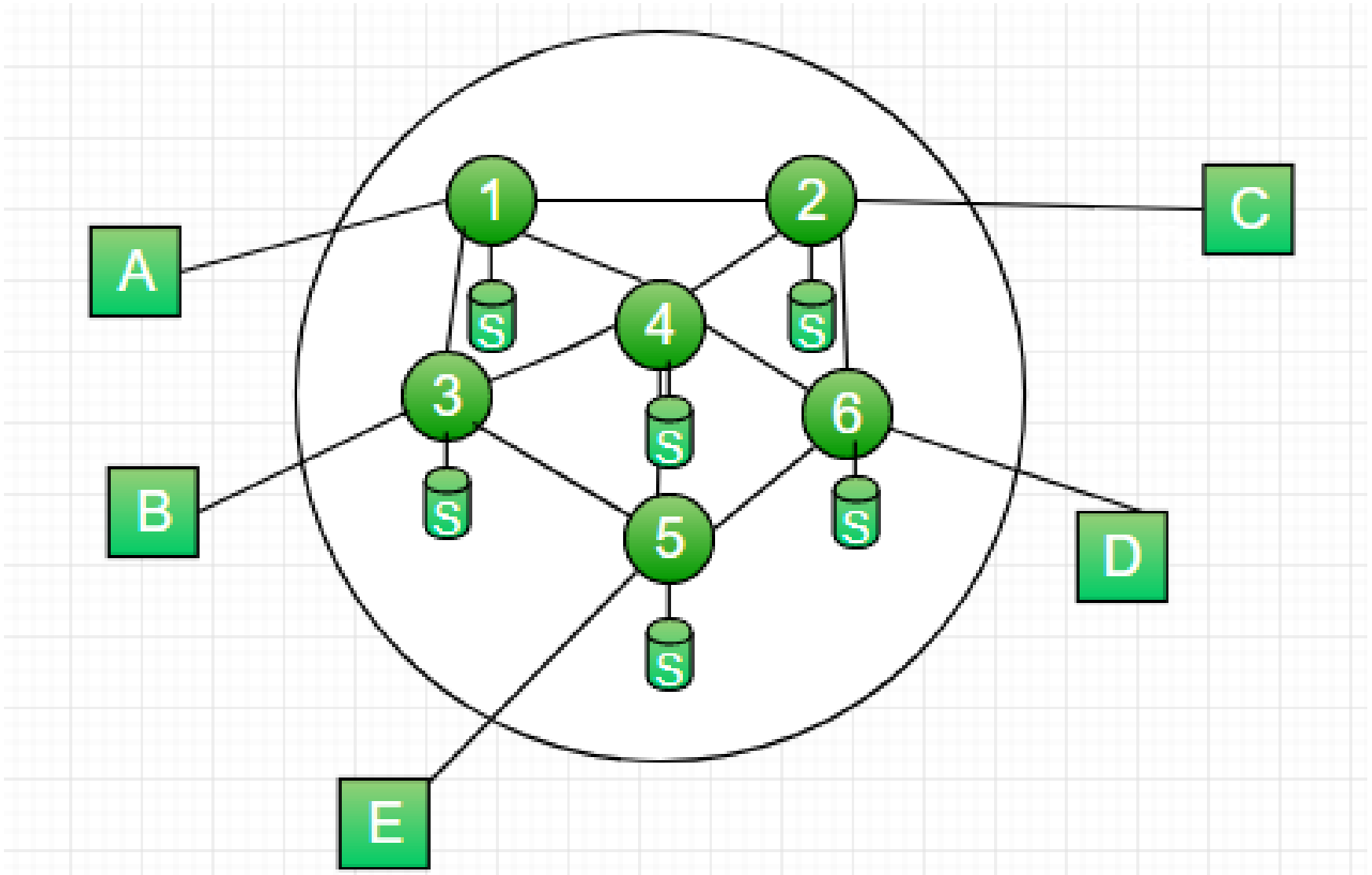


Figure 5-7.
The first five steps used in computing the shortest path from A to D. The arrows indicate the working node.

FLOODING:

A simple local technique in which every incoming packet is sent out on every outgoing line except the one it arrived on is called flooding.

- **Flooding generates large number of duplicate packets.** One way to present this is for each node to renumber the identity of the packets it has already sent.
- One such measure is to have a hop counter contained in the header of each packet, **which is decremented at each hop. When the count reaches zero, the packet is discarded**. The counter is set to maximum value i.e diameter of subnet.
- **No** routing table maintained.
- Always gives shortest path.
- It is more reliable.
- Traffic is high.
- Duplicate packets are present.


A
B
C
D
E
1
2
3
4
5
6
S
S
S
S
S
S

Characteristics –

•All possible routes between Source and Destination is tried.

•A packet will always get through if path exists

•As all routes are tried, there will be at least one route which is the shortest

•All nodes directly or indirectly connected are visited.

Limitations –

•Flooding generates vast number of duplicate packets.

•Suitable damping mechanism must be used.

Hop-Count –

•A hop counter may be contained in the packet header which is decremented at each hop.

with the packet being discarded when the counter becomes zero.

•The sender initializes the hop counter. If no estimate is known, it is set to the full diameter of the subnet.

•Keep track of the packets which are responsible for flooding using a sequence number. Avoid sending them out a second time.

Hierarchical Routing:

* **When the network size grows the number of routers in the network increases.**

* Consequently the **size of routing tables as well and routers cannot handle network traffic as efficiently.**

* We use hierarchical routing to overcome this problem.

    * Hierarchical Routing is essentially **Divide and Conquer strategy,**

* The **network is divided into different regions and a router for a particular region knows only about its own domain and other neighbor routers.**

    * In **hierarchical routing routers are classified in to groups known as regions.**

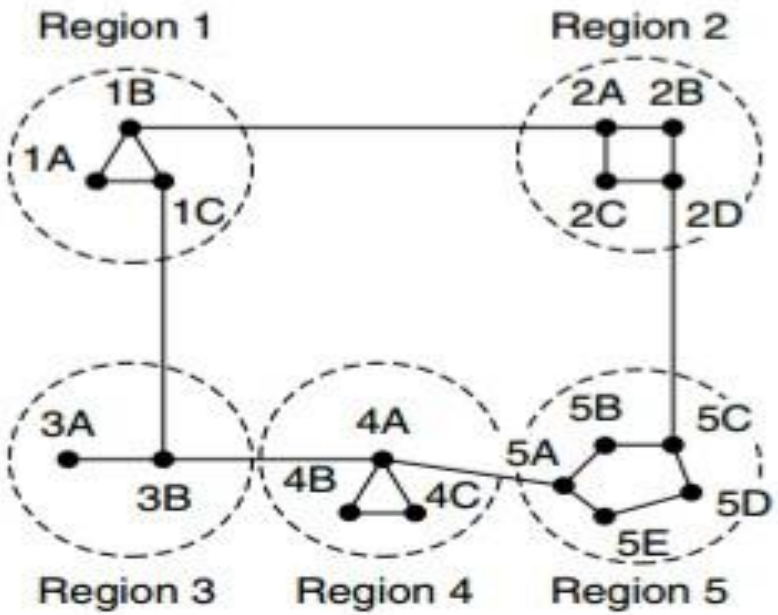
* **Each router has only the information about the routers in its own region and has no information about routers in the other regions** .

* **So routers just save one record in their table for every other region.**

- **Router just save one record in their table for every other region.**
- **Router only contains the record of their immediate neighbors in the table.**
- **In three level hierarchical routing the network is called into a number of clusters.**
- **Each cluster is made of a number of regions and each region contains a number or routers.**
- **In this method it will route first to the region then to the IP Prefix within the region- hide details within a region from outside of the region.**

In three-level hierarchical routing, the network is classified into a number of **clusters**. Each cluster is made up of a number of regions, and each region contains a number or routers. Hierarchical routing is widely used in Internet routing and makes use of several routing protocols.





(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Figure 5-14. Hierarchical routing.

In three-level hierarchical routing, the network is classified into a number of **clusters**. Each cluster is made up of a number of regions, and each region contains a number or routers. Hierarchical routing is widely used in Internet routing and makes use of several routing protocols.

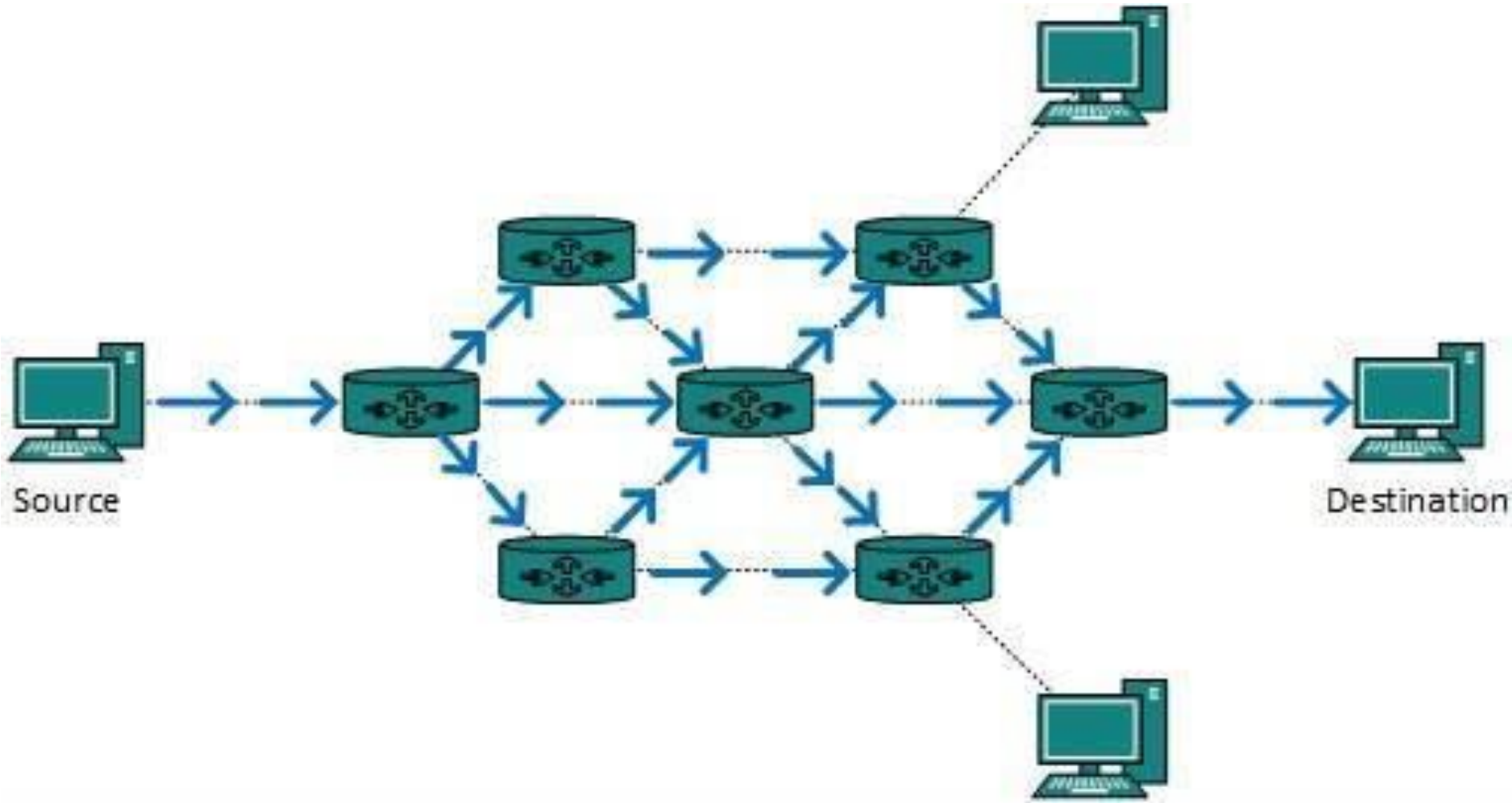


Broadcast Routing:

Sending a packet to all destinations simultaneously is called broadcasting.

Ex: a service distributing weather reports, stock market updates, or live radio programs might work best by sending to all machines and letting those that are interested read the data.

- Various methods are proposed but some are not in practice. for **example** sending the data to each destination in a network which is widely applicable but wasteful of bandwidth and source must have the complete list of all destinations.
- **Multi destination routing**: In multi-destination routing, each packet contains either a list of destinations. When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed. **This is also not made in practice because of bandwidth and destination list problems.**


Source
Destination

- The idea for **reverse path forwarding** is elegant and remarkably simple once it has been pointed out. When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets toward the source of the broadcast. If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate otherwise it is accepted.

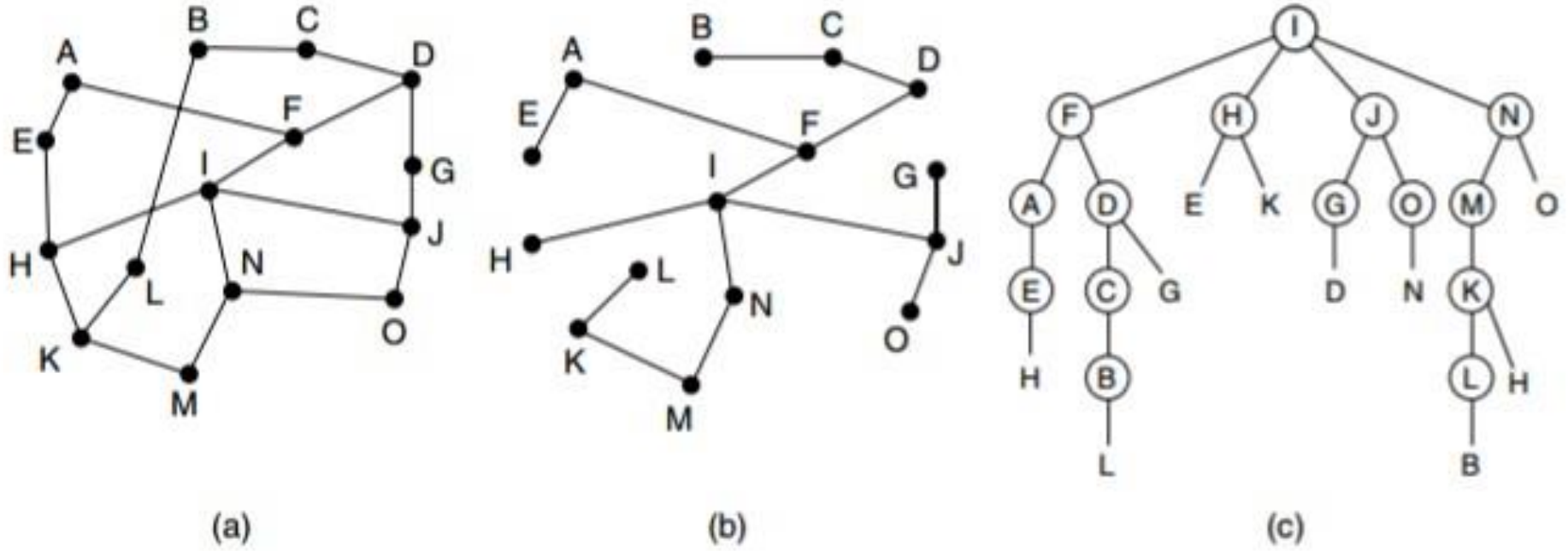


Figure 5-15. Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

Multicast Routing:

- **Sending a message to well-defined groups** that are numerically large in size but small compared to the network as a whole is called multicasting, and the routing algorithm used is called multicast routing.
- **All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group.** How these tasks are accomplished is not of concern to the routing algorithm.

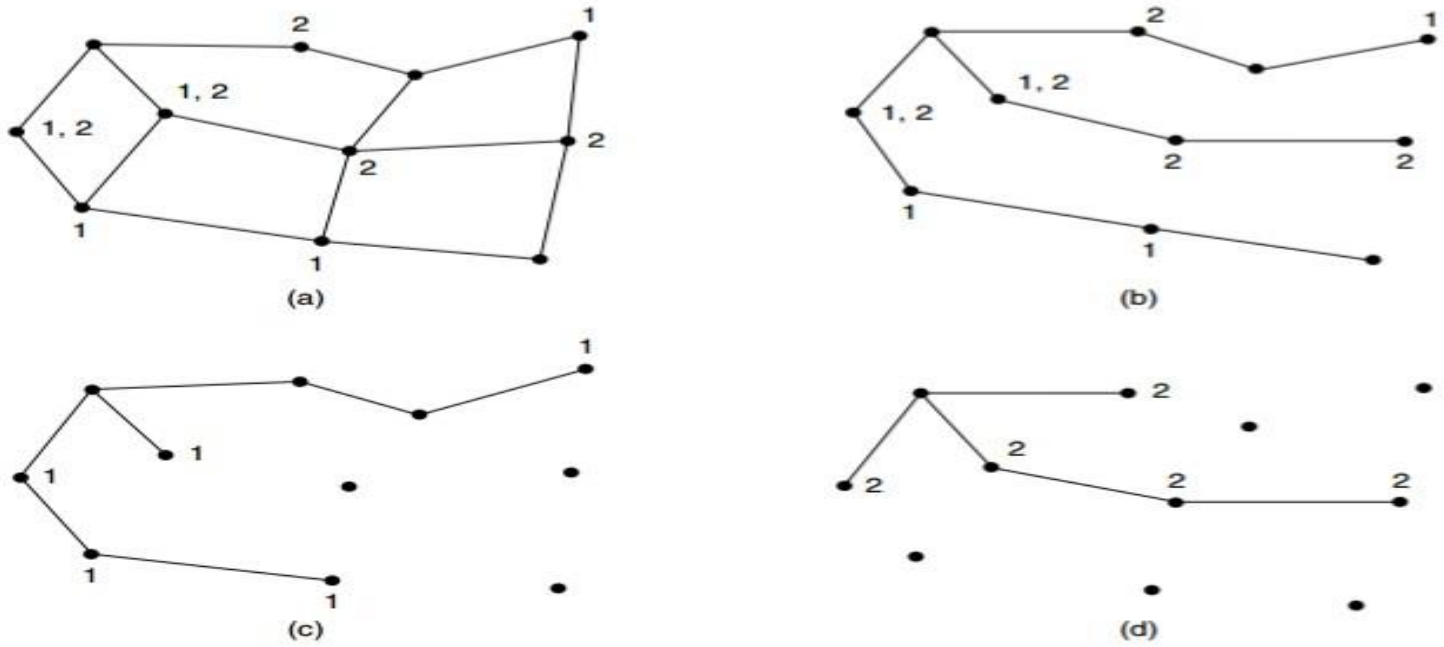


Figure 5-16. (a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

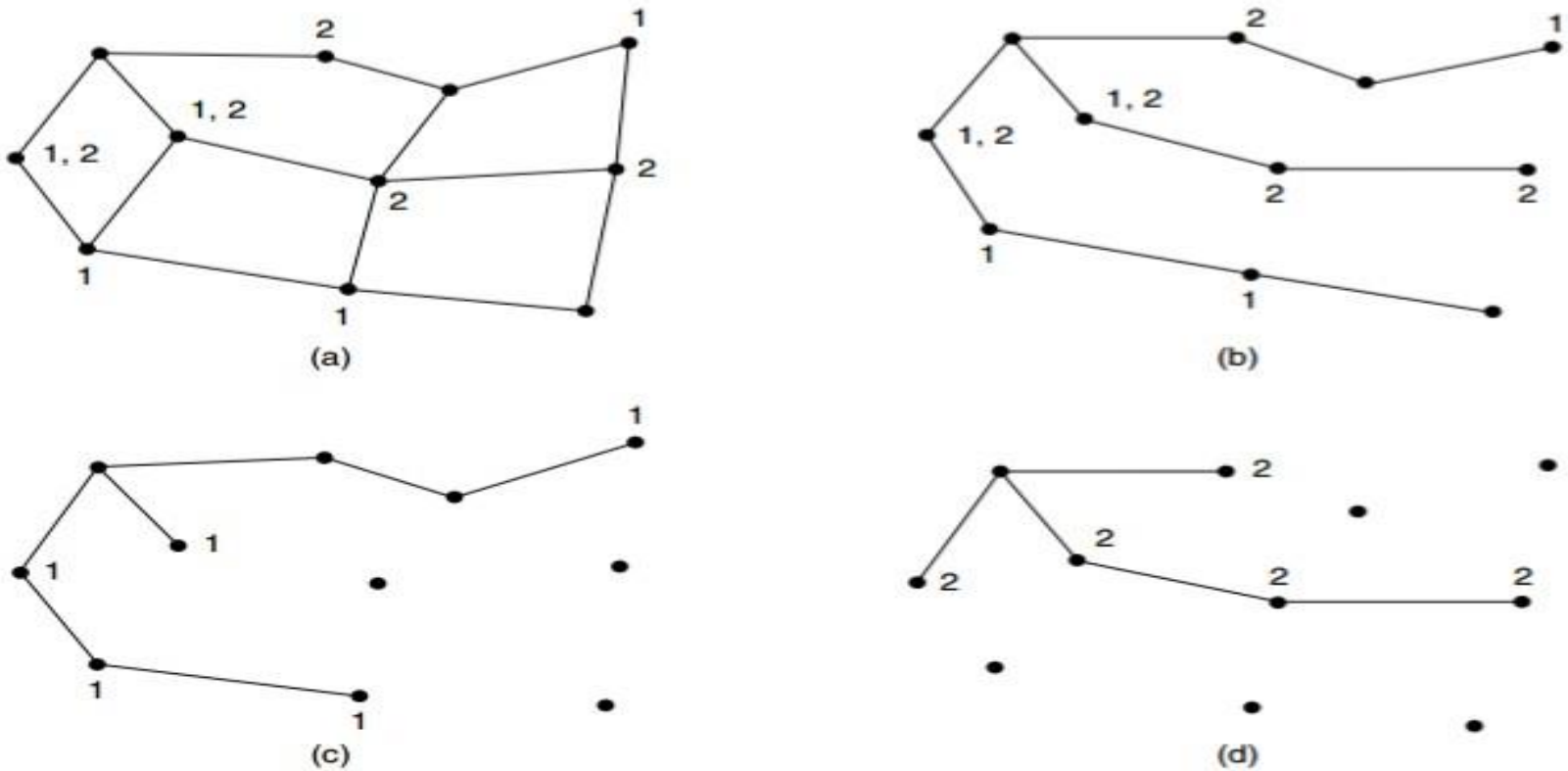


Figure 5-16. (a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

Example:

Consider the two groups, 1 and 2, in the network shown in Fig. 5-16(a). Some routers are attached to hosts that belong to one or both of these groups, as indicated in the figure.

A spanning tree for the leftmost router is shown in Fig. 5-16(b). This tree can be used for broadcast but is overkill for multicast, as can be seen from the two pruned versions that are shown next.

In Fig. 5-16(c), all the links that do not lead to hosts that are members of group 1 have been removed. The result is the multicast spanning tree for the leftmost router to send to group 1. Packets are forwarded only along this spanning tree, which is more efficient than the broadcast tree because there are 7 links instead of 10. Fig. 5-16(d) shows the multicast spanning tree after pruning for group 2. It is efficient too, with only five links this time. It also shows that different multicast groups have different spanning trees.

Link State Routing Algorithm:

Link state routing is the second family of routing protocols.

While distance vector routers use a distributed algorithm to compute their routing tables,

link-state routing uses link-state routers to exchange messages that allow each router to learn the entire network topology.

Based on this learned topology, each router is then able to compute its routing table by using a shortest path computation.

The idea behind link state routing is simple and can be stated as five parts.

Each router must do the following:

1. Discover its neighbors and learn their network addresses.
2. Measure the delay or cost to each of its neighbors.
3. Construct a packet telling all it has just learned.
4. Send this packet to all other routers.
5. Compute the shortest path to every other router.

- Fig :(a) shows a network, part (b) shows a sink tree for router I of that network, and part (c) shows how the reverse path algorithm works. On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree. Each of these packets arrives on the preferred path to I and is so indicated by a circle around the letter. On the second hop, eight packets are generated, two by each of the routers that received a packet on the first hop. As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line. Of the six packets generated on the third hop, only three arrive on the preferred path (at C, E, and K); the others are duplicates. After five hops and 24 packets, the broadcasting terminates, compared with four hops and 14 packets had the sink tree been followed exactly.

- . A **spanning tree(it covers all routers not cover all paths)(removing duplications)** is a subset of the network that includes all the routers but contains no loops. Sink trees are spanning trees. If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on. This method makes excellent use of bandwidth, generating the absolute minimum number of packets necessary to do the

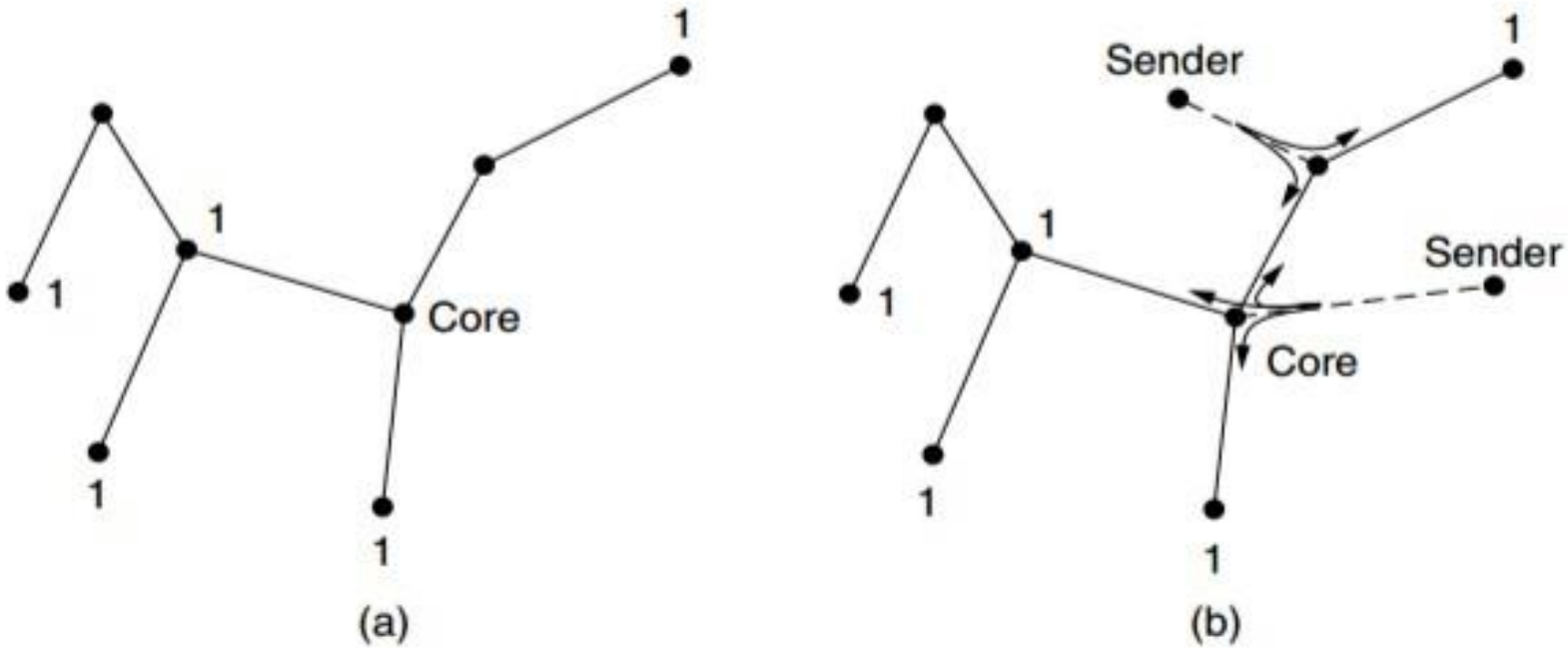


Figure 5-17. (a) Core-based tree for group 1. (b) Sending to group 1.

- An alternative design uses core-based trees to compute a single spanning tree for the group. All of the routers agree on a root (called the core or rendezvous point) and build the tree by sending a packet from each member to the root. The tree is the union of the paths traced by these packets. Fig. 5-17(a) shows a core-based tree for group 1. To send to this group, a sender sends a packet to the core. When the packet reaches the core, it is forwarded down the tree. This is shown in Fig. 5-17(b) for the sender on the right hand side of the network. As a performance optimization, packets destined for the group do not need to reach the core before they are multicast. As soon as a packet reaches the tree, it can be forwarded up toward the root, as well as down all the other branches. This is the case for the sender at the top of Fig. 5-17(b).
- Having a shared tree is not optimal for all sources. Shared trees can be a major savings in storage costs, messages sent, and computation. Each router has to keep only one tree per group, instead of m trees. Further, routers that are not part of the tree do no work at all to support the group. For this reason, shared tree approaches like core-based trees are used for multicasting to sparse groups in the Internet as part of popular protocols such as PIM (Protocol Independent Multicast).

Distance-vector routing protocol :(Bell man ford algorithm)

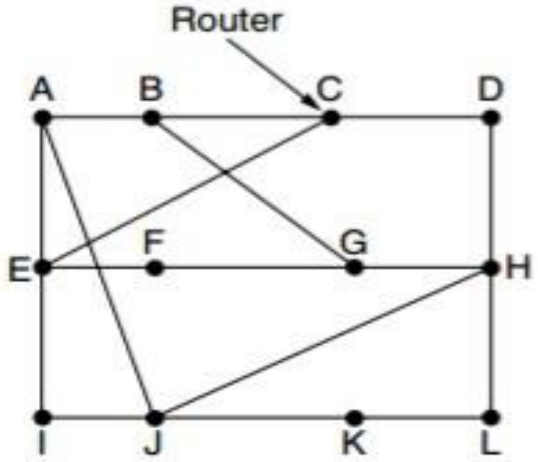
Distance-vector routing protocol in data networks determines the best route for data packets based on distance.

- Distance-vector routing protocols measure the distance by the number of routers a packet has to pass, one router counts as one hop.
- Some distance-vector protocols also take into account network latency (Best amount of time to transmission of data) and other factors that influence traffic on a given route.
- To determine the best route across a network, routers, on which a distance-vector protocol is implemented, exchange information with one another, usually routing tables plus hop counts for destination networks and possibly other traffic information.
- Distance-vector routing protocols also require that a router informs its neighbors of network topology changes periodically.

Distance-vector routing protocols use the Bellman–Ford algorithm and Ford–Fulkerson algorithm to calculate the best route.

Another way of calculating the best route across a network is based on link cost, and is implemented through link-state routing protocols.

- In distance vector routing, each router maintains a routing table indexed by, and containing one entry for each router in the network. This entry has two parts: the preferred outgoing line to use for that destination and an estimate of the distance to that destination.



(a)

To	A	I	H	K	New estimated delay from J	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K
	JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6	New routing table for J	

Vectors received from J's four neighbors

(b)

Figure 5-9. (a) A network. (b) Input from *A*, *I*, *H*, *K*, and the new routing table for *J*.

Example: JA = 8

JI = 10

JH =12

JK = 6

- J ----> A

JA=8

JI + IA = 10 +24 =34

JH + HA = 12 +20 =32

JK + KA =6 + 21 =27

- J -----> B

JA + AB =8 + 12 =20

JI + IB = 10 +36 =46

JH + HB = 12 + 31 =43

JK + KB = 6+ 28 = 34

Example: JA = 8

JI = 10

JH =12

JK = 6

- J ----> A

JA=8

JI + IA = 10 +24 =34

JH + HA = 12 +20 =32

JK + KA =6 + 21 =27

- J -----> B

JA + AB =8 + 12 =20

JI + IB = 10 +36 =46

JH + HB = 12 + 31 =43

JK + KB = 6+ 28 = 34

Link State Routing Algorithm:

Link state routing is the second family of routing protocols.

While distance vector routers use a distributed algorithm to compute their routing tables,

link-state routing uses link-state routers to exchange messages that allow each router to learn the entire network topology.

Based on this learned topology, each router is then able to compute its routing table by using a shortest path computation.

The idea behind link state routing is simple and can be stated as five parts.

Each router must do the following:

1. Discover its neighbors and learn their network addresses.
2. Measure the delay or cost to each of its neighbors.
3. Construct a packet telling all it has just learned.
4. Send this packet to all other routers.
5. Compute the shortest path to every other router.

Features of link state routing protocols –

- **Link state packet –** A small packet that contains routing information.
- **Link state database –** A collection information gathered from link state packet.
- **Shortest path first algorithm (Dijkstra algorithm) –** A calculation performed on the database results into shortest path
- **Routing table –** A list of known paths and interfaces.

Calculation of shortest path –

To find shortest path, each node need to run the famous **Dijkstra algorithm**. This famous algorithm uses the following steps:

Step-1: The node is taken and chosen as a root node of the tree, this creates the tree with a single node, and now set the total cost of each node to some value based on the information in Link State Database

Step-2: Now the node selects one node, among all the nodes not in the tree like structure, which is nearest to the root, and adds this to the tree.The shape of the tree gets changed .

Step-3: After this node is added to the tree, the cost of all the nodes not in the tree needs to be updated because the paths may have been changed.

Step-4: The node repeats the Step 2. and Step 3. until all the nodes are added in the tree

Congestion Control in Computer Networks:

What is **congestion**?

A state occurring in network layer when the message traffic is so heavy that it slows down network response time.

(or)

When too many packets are present in (a part of) the subnet, performance degrades. This situation is called congestion

- **Congestion control** refers to the techniques used to control or prevent congestion.

- **Congestion control** is the process of maintaining the number of packets in the network below a certain level at which performance falls off.

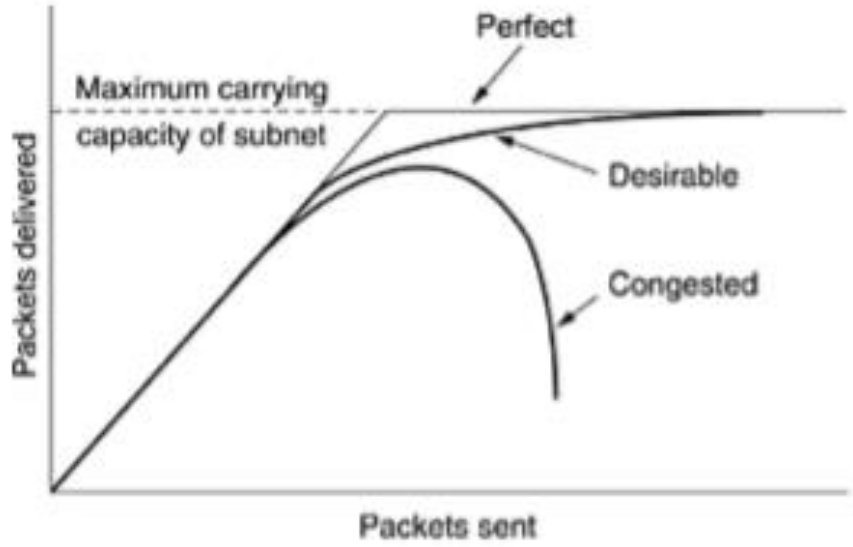
Effects of Congestion

•As delay increases, performance decreases.
•If delay increases, retransmission occurs, making situation worse.

When the number of packets dumped into the subnet by the hosts is within its carrying capacity, they are all delivered (except for a few that are afflicted with transmission errors) and the **number delivered is proportional to the number sent.**

However, as **traffic increases too far**, the routers are no longer able to cope and **they begin losing packets.** This tends to make matters **worse**. At **very high trafffic, performance collapses completely and almost no packets are delivered.**

When too much traffic is offered, congestion sets in and performance degrades sharply.



Causes Of Congestion : (Congestion can be brought on by several factors)

- If all of a sudden, **streams of packets begin arriving on three or four input lines and all need the same output line**, a queue will build up.

- If there is insufficient memory to hold all of them, packets will be lost.

- **Slow processors can also cause congestion**. If the routers' CPUs are slow at performing the bookkeeping tasks required of them (queueing buffers, updating tables, etc.), queues can build up, even though there is excess line capacity.

- Similarly, low-bandwidth lines can also cause congestion. Upgrading the lines but not changing the processors, or vice versa.

- The band width of the links are also important in congestion . The links to be used must be high bandwidth to avoid the congestion.

- When the arrival of the packets are not uniform i.e when the traffic is bursty.

Congestion Control Algorithms – Easy Point-wise Notes

1. Congestion

Congestion means **too many packets are present in the network**.
It causes **packet delay and packet loss**.
It reduces the **performance of the network**.
Congestion occurs when the **traffic is more than the network capacity**.

2. Congestion Control

Congestion control manages **heavy traffic in the network**.
It prevents the network from becoming overloaded.
It involves **hosts and routers**.
The sender reduces its transmission rate when congestion occurs.

3. Flow Control vs Congestion Control

Flow Control

It controls the data flow between a **sender and receiver**.
It prevents a fast sender from sending data too quickly.
It mainly protects the **receiver**.

Congestion Control

It controls the traffic in the **entire network**.
It prevents too many packets from entering the network.
It mainly protects the **network**.

Easy difference:

Flow Control → Protects the receiver.
Congestion Control → Protects the network.

Congestion Control Techniques

1. Warning Bit

The router detects congestion.
It sets a **warning bit** in the packet.
The warning bit is sent back to the sender through an ACK.
The sender checks the warning bit.
The sender then **reduces its transmission rate**.

2. Choke Packets

A choke packet is a **control packet**.
It is generated by a congested router.
It is sent to the source.
It tells the source to **slow down**.
The source reduces its transmission rate.

Hop-by-Hop Choke Packets

The choke message is sent **hop by hop**.
Each router reduces its transmission rate.
It helps to control congestion faster.
It is useful for **long-distance networks**.

3. Load Shedding

Load shedding means **discarding packets**.
Routers discard packets when their buffers are full.
The router decides which packets to discard.
For file transfer, important packets should not be discarded unnecessarily.
For voice and video, old packets can be discarded.
Packets can be given **discard priority**.

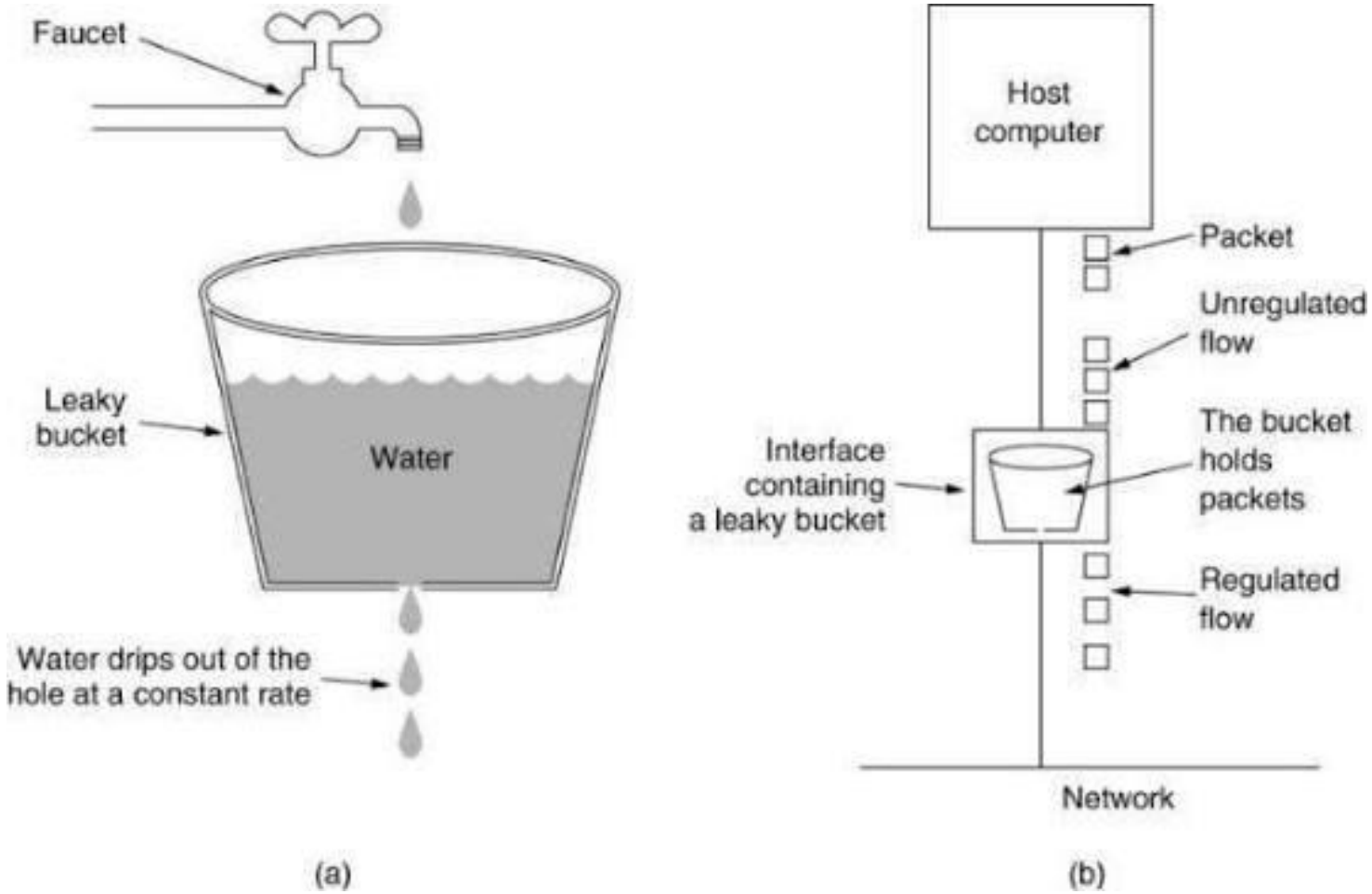
5. Traffic Shaping

Traffic shaping controls the **rate of packet transmission**.
It controls packets before they enter the network.
It reduces sudden bursts of traffic.
It helps to reduce congestion.
Two important traffic-shaping algorithms are:

Leaky Bucket

Example: Leaky Bucket Algorithm

Leaky Bucket is used to **control network traffic**.
Packets are stored in a buffer called a bucket.
Packets leave the bucket at a **constant rate**.
It converts bursty traffic into smooth traffic.
If the bucket becomes full, new packets are **discarded**.
It does not allow large bursts of packets.
It provides a **uniform output rate**.


Faucet
Leaky bucket
Water
Water drips out of the hole at a constant rate
(a)
Host computer
Packet
Unregulated flow
The bucket holds packets
Interface containing a leaky bucket
Regulated flow
Network
(b)

General Principles of Congestion Control :

This approach leads to dividing all solutions into two groups:

- **Open loop Congestion Control.**
- **Closed loop Congestion Control.**

Open Loop Congestion Control :

Open loop congestion control policies are applied to prevent congestion before it happens. The congestion control is handled either by the source or the destination.

Closed Loop Congestion Control:

Closed loop congestion control technique is used to treat or alleviate congestion after it happens. Several techniques are used by different protocols.

This approach has three parts when applied to congestion control:

1. **Monitor the system to detect when and where congestion occurs.**
2. **Pass this information to places where action can be taken.**
3. **Adjust system operation to correct the problem.**

Congestion Prevention Policies :

Figure 5-26. Policies that affect congestion.

Layer	Policies
Transport	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy • Timeout determination
Network	• Virtual circuits versus datagram inside the subnet • Packet queueing and service policy • Packet discard policy • Routing algorithm • Packet lifetime management
Data link	• Retransmission policy • Out-of-order caching policy • Acknowledgement policy • Flow control policy

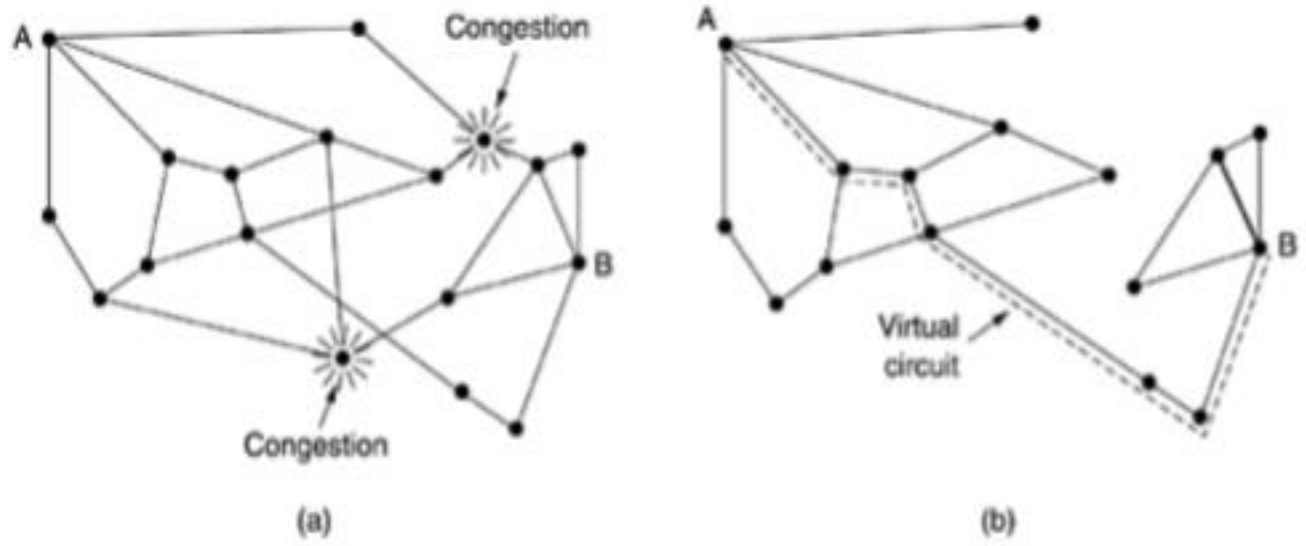
Congestion Control in Virtual-Circuit Subnets:

The congestion control methods described above are basically open loop:

They try to prevent congestion from occurring in the first place, rather than dealing with it after the fact. In this section we will describe some approaches to dynamically controlling congestion in virtual circuit subnets. In the next two, we will look at techniques that can be used in any subnet.

One technique that is widely used to keep congestion that has already started from getting worse is admission control. The idea is simple: **once congestion has been signaled, no more virtual circuits are set up until the problem has gone away. Thus, attempts to set up new transport layer connections fail.**

Figure 5-27. (a) A congested subnet. (b) A redrawn subnet that eliminates the congestion. A virtual circuit from A to B is also shown



Congestion Control in Datagram Subnets:

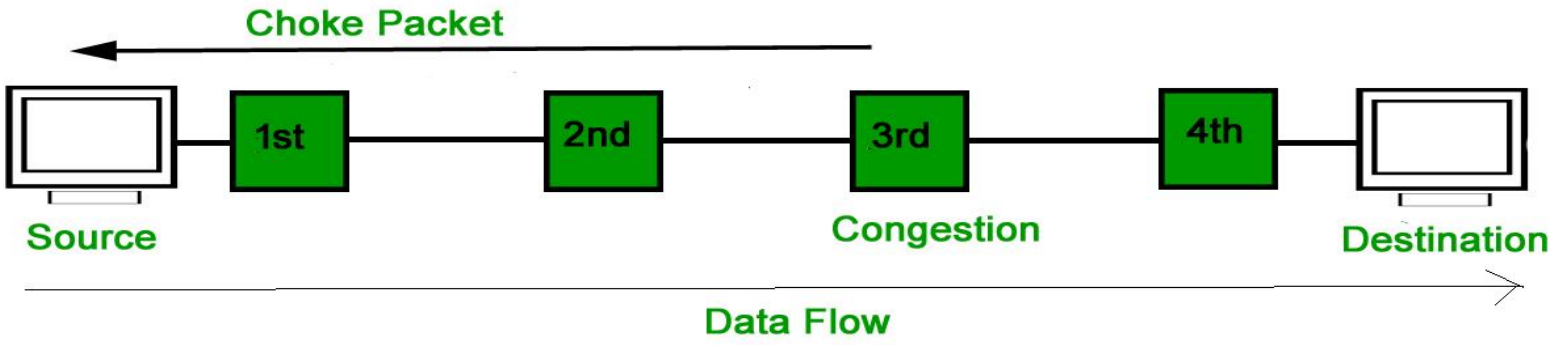
some approaches that can be used in datagram subnets (and also in virtual circuit subnets). Each router can easily monitor the utilization of its output lines and other resources.

Warning Bit :

- **It is Special bit in the packet header is set by the router to warn the source when congestion is detected.**
- **The bit is copied and Ack is send to the sender.**
- **The sender monitors the number of Ack packets it receives with the warning bit set and adjust its transmission rate accordingly.**

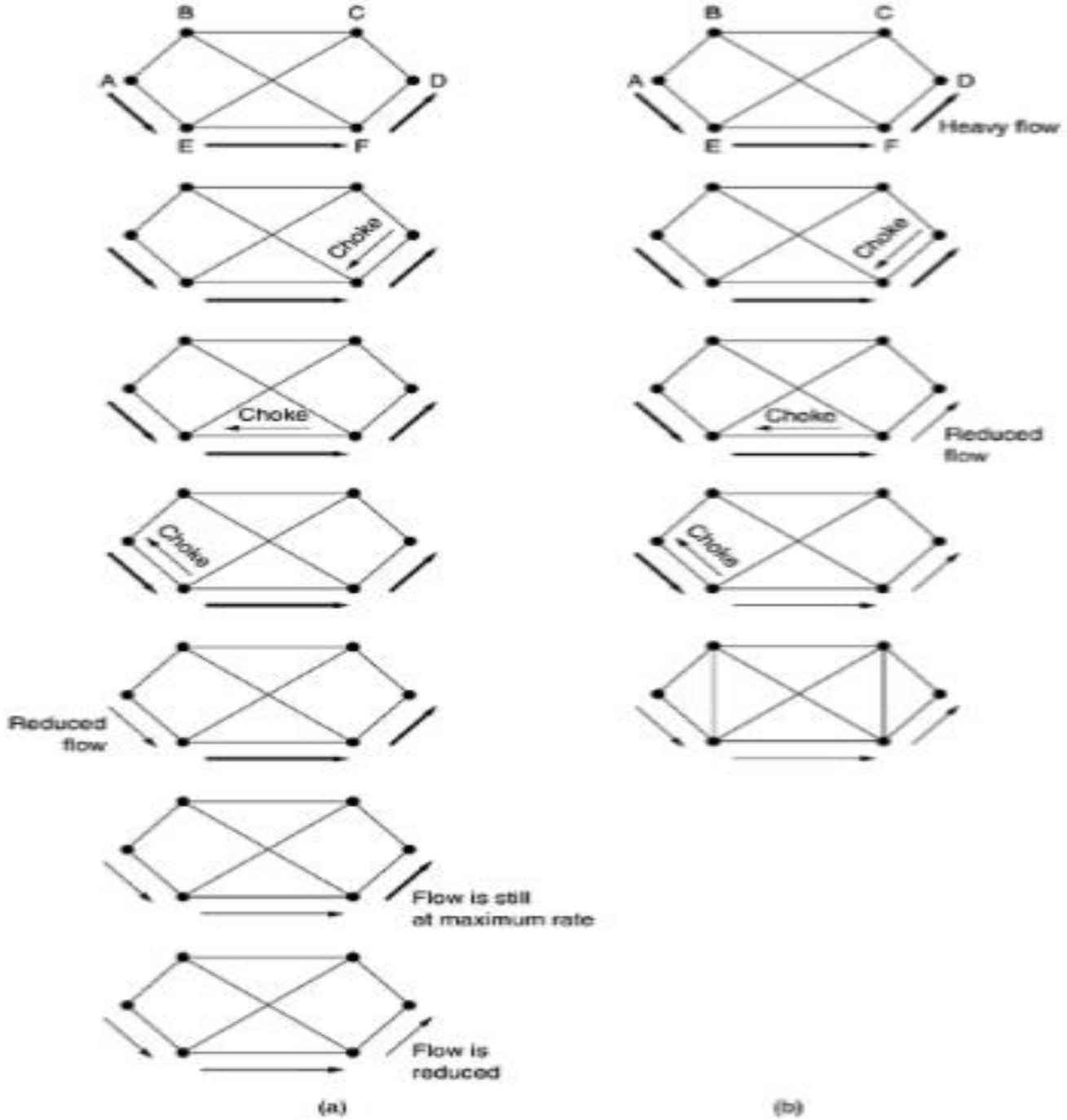
Choke Packets :

A **choke packet** is a **packet** sent by a node to the source to inform it of **congestion**. Each router monitor its resources and the utilization at each of its output lines. ... The intermediate nodes through which the **packets** has traveled are not warned about **congestion**.



Hop-by-Hop Choke Packets :

At high speeds or over long distances, sending a choke packet to the source hosts does not work well because the reaction is so slow.



Hop-by-Hop Choke Packets :

At high speeds or over long distances, sending a choke packet to the source hosts does not work well because the reaction is so slow.

Consider, for example, a host in San Francisco (router A in Fig. 5-28) that is sending traffic to a host in New York (router D in Fig. 5-28) at 155 Mbps. If the New York host begins to run out of buffers, it will take about 30 msec for a choke packet to get back to San Francisco to tell it to slow down. The choke packet propagation is shown as the second, third, and fourth steps in Fig. 5-28(a). In those 30 msec, another 4.6 megabits will have been sent. Even if the host in San Francisco completely shuts down immediately, the 4.6 megabits in the pipe will continue to pour in and have to be dealt with. Only in the seventh diagram in Fig. 5-28(a) will the New York router notice a slower flow.

Figure 5-28. (a) A choke packet that affects only the source. (b) A choke packet that affects each hop it passes through.

Quality of service:

- We have to provide good quality of service basing on the demand of applications (and customers), even though we have provided algorithms that reduced congestion and improved network performance.
- Multimedia applications in particular, often need a **minimum throughput** and **maximum latency** to work.
- An easy solution to provide good quality of service is to build a network with enough capacity for whatever traffic will be thrown at it and we call it as **overprovisioning**. The resulting network will carry application traffic without significant loss and, assuming a decent routing scheme, will deliver packets with low latency
- Overprovisioning is expensive to handle. It is based on expected traffic. All bets (expectations) are off if the traffic changes too much.
- Quality of service let a network with less capacity meet application requirements just as well at a lower cost. With quality of service, the network can honor the performance guarantees that it makes even when traffic spikes, at the cost of turning down some requests.

- Four issues must be addressed to ensure quality of service:

1. What applications need from the network?
2. How to regulate the traffic that enters the network.
3. How to reserve resources at routers to guarantee performance.
4. Whether the network can safely accept more traffic.

Several common applications and the stringency of their network requirements are listed below:

Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low